

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM AND LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

growing large-area, single-crystal, wafer-scale graphene. Engineers around the world have invested heavily in research for the commercialisation of graphene, but have faced many obstacles due to the challenges associated with it. The new method synthesises large-area graphene into a single crystal on a semiconductor, maintaining its electric and mechanical properties.

New lithium battery created in Japan

The long life of lithium-ion batteries makes them the rechargeable choice for everything from implantable medical devices to wearable consumer electronics. But lithium-ion batteries rely on liquid chemistries involving lithium salts dissolved in organic solvents, creating flame risks that would be avoided if the cells were completely solid state.

Now a team of researchers at Tohoku University in Japan has created a new type of lithium-ion conductor for future batteries that could be the basis for a whole new generation of solid-state batteries. It uses rock salt lithium borohydride (LiBH₄), a well-known agent in organic chemistry laboratories that has been considered for batteries before, but up to now it has only worked at high temperatures or pressures.

In the journal APL Materials, from AIP Publishing, the researchers describe how they doped a cubic lattice of KI molecules with the LiBH₄. This allowed them to stabilise the high-pressure form of lithium borohydride and make a solid solution at normal atmospheric pressure that was stable at room temperature.

Gesture recognition and virtual touch for vehicle controls

A new automotive cockpit concept demonstrates that futuristic-advancements, such as 3D gesture recognition, are now closer to reality. The Horizon cockpit concept, which Visteon has been demonstrating to global vehicle manufacturers, blends three emerging technologies to transform the way a driver controls features such as interior temperature, audio and navigation.

Gesture control, an advanced camera system, maps the user's hand to replicate a virtual hand on the centre stack. Horizon drivers can then easily operate certain vehicle controls simply by moving their hand or just a finger, without making contact with the instrument cluster. For example, radio volume can be adjusted by making a turning motion with one's hand.

By integrating a pressure-sensitive touch pad with a virtual touch screen, drivers can operate centre-stack controls without having to physically reach for them. The touch pad recognises the amount of pressure applied for improved responses. The touch pad can be implemented with any soft material, such as leather or cloth, allowing flexibility for its location.

High-resolution graphics present information on two separate planes, bringing only those controls with which the driver is interacting to the forefront. The driver can virtually 'push' through the graphics on the first pane to access the second.